

Chapter 11: Nonequilibrium (Classical):

Exercise #1: Boltzmann statistics is about individual particles in a system.
(pg 635)

Each particle contributes to the total distribution by a general function, $f(x) = N \cdot p(x)$, where $N = \#$ of particles and $p(x) =$ particle distribution. However, Boltzmann statistics never incorporate strong correlations between particles.

Exercise #2: (11.4) "Langevin Correlation"

(pg 644)

$$\langle \xi_i(t), \xi_{i'}(t') \rangle = A \delta_{ii'} \delta(t-t')$$

(pg 643) "Gaussian Functional Distribution"

$$P(\xi) = N e^{-\frac{\delta \Delta t}{2A} |\xi|^2}$$

$$\langle \xi(\Delta t) \xi_i(\Delta t) \rangle = \int_{-\infty}^{\infty} \xi_i(t) \cdot e^{-\frac{\delta \Delta t}{4A} |\xi|^2} d\xi$$

$$= - \left(\frac{\xi}{\frac{2(\delta\Delta t)}{4A}} \right) e^{\frac{-\delta\Delta t}{4A} |\xi|^2} \Big|_{-\infty}^{\infty}$$

$$+ \frac{1}{2(\delta\Delta t)/4A} \int_{-\infty}^{\infty} e^{\frac{-\delta\Delta t}{4A} |\xi|^2} d\xi$$

$$= \sqrt{\pi} \left(\frac{A}{\delta\Delta t} \right)^{3/2}$$

Exercise #3: (Pg 645) "Langevin Equation for overdamped limit"
(Pg 645)

$$m \gamma \partial_t q = \xi$$

(11.15) "Fokker-Planck Equation"

$$(\partial_t - \partial_{v_i} \gamma v_i - \partial_{v_i v_i}^2 D_v) p(v, t) = 0$$

$$\text{where } D_v = \frac{A}{2m^2} = \gamma^2 D$$

Exercise #4:
(pg 645)

(from exercise) "Generalized phase space
Langevin Equation"

$$d_t \bar{q} = \frac{\partial H}{\partial p}$$

$$d_t p = -\frac{\partial H}{\partial q} - \gamma p + \xi$$

$$= -\frac{\partial p}{\partial t} - \gamma p + m \gamma d_t q$$

$$= -\frac{\partial p}{\partial t} - \gamma p + \frac{D_p}{T} d_t q$$

$$= -\frac{\partial p}{\partial t} - \gamma p + \frac{D_p}{T} \frac{\partial H}{\partial p}$$

$$= -\frac{\partial p}{\partial t} - \gamma p - \frac{D_p}{p}$$

$$= \dots$$

$$= -\frac{\partial H}{\partial q} + \frac{\partial}{\partial p} \left(\gamma p + D_p \frac{\partial}{\partial p} \right) p(x, t)$$

$$= -\{H, p(x, t)\}$$

Exercise #5:
(pg 650)

(pg 650) "Particle energy, and
momentum density"

$$\left. \begin{array}{l} p(q, t) \\ E(q, t) \\ \pi(q, t) \end{array} \right\} = \frac{1}{\delta V} \int_{\delta T} F(x, t) \left\{ \begin{array}{l} H(x) \\ p \end{array} \right.$$

(11.23) "Energy Conservation"

$$\partial_t E + \nabla \cdot j_e = 0$$

(from problem) "Current density"

$$j_e \equiv \frac{1}{\delta V} \int_{\delta T} H \cdot v$$

$$\partial_t E + \nabla \cdot j_e = \frac{\partial}{\partial t} \frac{1}{\delta V} \int f(x,t) \cdot H(x) + \frac{1}{\delta V} \int_{\delta T} H \cdot v$$

$$= 0$$

(from) "problem" "Momentum current density"

$$\pi_{\alpha\beta} = \frac{1}{\delta V} \int_{\delta T} f \cdot m \cdot v_\alpha \cdot v_\beta$$

(11.23) "Momentum Conservation"

$$\partial_t \pi_\alpha + \partial_{q_\beta} \pi_{\alpha\beta} = \frac{\partial}{\partial t} \frac{1}{\delta V} \int f(x,t) \cdot p + \frac{\partial}{\partial q} \frac{1}{\delta V} \int_{\delta T} f \cdot m \cdot v_\alpha \cdot v_\beta$$

$$= 0$$

The author provided integrals with no variable of integration. Potato, potato.

Exercise #6: (from exercise) "Master Equation"

(Pg 657)

$$\partial_t p(a, t) = \int da' [W(a|a')p(a'|t) - W(a'|a)p(a, t)]$$

with $W(a|a') = u(a) \cdot v(a')$

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FACTORIZABLE
TRANSITION
MATRIX

(from exercise) "Laplace Transform"

$$p(a, z) = \int_0^\infty dt e^{izt} \cdot p(a, t)$$

$$\partial_t p(a, z) = -p(a, t=0) - izp(a, z)$$

$$= \int da' [W(a|a')p(a'|t) - W(a'|a)p(a, t)]$$

$$= \int da' [u(a)v(a')p(a'|z) - u(a)v(a')p(a|z)]$$

$$= u(a) \int da' v(a')p(a'|z) - v(a)p(a|z) \int da' u(a')$$

$$-p(a, t=0) - izp(a, z) = u(a) \int da' v(a')p(a'|z) - v(a)p(a|z) \int da' u(a')$$

$$p(a, z)(v(a)/\tau - iz) = p(a, t=0) + u(a)\sigma(z)$$

$$p(a, z) = \frac{p(a, t=0) + u(a)\sigma(z)}{-i z + v(a)/\tau}$$

$$\text{when } 1/\tau = \int da u(a), \sigma(z)$$

$$\sigma(z) = \int da v(a) p(a, z)$$

$\sigma(z)$ solution:

$$\sigma(z) = \int da v(a) \cdot p(a|z)$$

$$= \int da v(a) \frac{p(a, t=0) + v(a)\sigma(z)}{-i z + v(a)/\tau}$$

$$= \int da \frac{v(a)p(a, t=0)}{-i z + v(a)/\tau} + \sigma(z) \int da \frac{u(a)}{-i z + v(a)/\tau}$$

$$\sigma(z) = \frac{\int da' \frac{v(a') p(a', t=0)}{-i z + v(a')/\tau}}{1 - \int da' \frac{u(a') v(a')}{-i z + v(a')/\tau}}$$

$$1 - \int da' \frac{u(a') v(a')}{-i z + v(a')/\tau}$$

Exercise #7:
(pg 658)

(from exercise) "Rate of change"

$$\dot{P}_n = r_{n+1} P_{n+1} + s_{n-1} P_{n-1} - (r_n + s_n) P_n$$

(from exercise) "Generating Function"

$$g(z, t) = \sum_{n=-\infty}^{\infty} z^n \cdot P_n(t) \quad \text{where } g(1, t) = 1$$

(from exercise) "inverse Mellin Transform"

$$p_n(t) = \frac{1}{2\pi i} \oint dz z^{-1-n} g(z, t)$$

$$= \frac{1}{2\pi i} \oint dz \frac{g(z, t)}{z^{n+1}}$$

$$= \frac{1}{2\pi i} \oint dz \frac{e^{(\frac{1}{z} + z - z)t}}{z^{n+1}}$$

(pg 404) "Cauchy Integral - modified"

$$C(w) = \frac{-1}{2\pi i} \int dw \frac{C^+(w')}{(w - w')} dz$$

Note: Book shows $n=0$ order Cauchy.

Wikipedia states n^{th} -term values.

$$F^{(n)}(w) = \frac{n!}{2\pi i} \oint \frac{f(w)}{(z - a)^{n+1}} dw$$

Note: The above equation has similarities to the inverse Mellin Transform.

a) If $r_n = s_n = 1$ and $p_n(0) = \delta_{n,0}$, then

$$\dot{g}(z, t) = \sum_{n=0}^{\infty} z^n |p_{n-1}(t) + p_{n+1}(t) - 2p_n(t)|$$

$$= z g(z, t) + \frac{1}{z} g(z, t) - 2g(z, t)$$

$$= (z + \frac{1}{z} - 2) g(z, t)$$

$$\text{where } g(z, t) = e^{(\frac{1}{z} + z - 2)t}$$

The second part has no immediate solution. "Saddle-point approximation" not in book.

b) If $r_n = 0$ and $s_n = \gamma_n$ and $p_n(0) = \delta_{n,0}$,

$$\text{then } \dot{g}(z, t) = \sum_{n=0}^{\infty} z^n |\gamma_n p_{n+1}(t) - 2p_n(t)|$$

$$\text{when } g(z, t) = e^{(\gamma_n z - z)t}$$

c) If $r_n = \alpha_n$ and $s_n = \beta_n$ and $p_n(0) = \delta_{n,0}$,

$$\text{then } \dot{g}(z, t) = \sum_{n=0}^{\infty} z^n |\alpha_n p_{n-1}(t) + \beta_n p_{n+1}(t) - 2p_n(t)|$$

$$\text{when } g(z, t) = e^{\left(\frac{\kappa \eta}{z} + z - z\right)t}$$

Exercise #7: (11.54) "Operators"

(pg 670)

$$\bar{a} = e^{\hat{p}} \hat{q} \quad a = e^{-\hat{p}}$$

(11.52) "States"

$$\langle n | \psi_n \rangle = \psi_n^n = e^{n \ln \psi_n}$$

$$\langle \psi_0 | n_0 \rangle = \frac{1}{n_0!} \bar{\psi}_0^{n_0} = e^{n_0 \ln \bar{\psi}_0 - \ln n_0!}$$

$$a | \psi \rangle = e^{\hat{p}} \hat{q} e^{n \ln \psi_n}$$

$$= 1 | \psi \rangle \quad \dots \text{if } \hat{q} = 1, \hat{p} = -i\hbar \frac{d}{dx}$$

$$a | \psi_0 \rangle = e^{-\hat{p}} e^{n_0 \ln \bar{\psi}_0 - \ln n_0!}$$

$$= 1 | \psi \rangle$$

Cole-Hopf Transformation:

(11.53) "Doi - Peliti Functional Integral"

$$P(n, t) = \int D\psi e^{-\int_0^t dt [-\partial_t \bar{\psi} \psi + H(\bar{\psi}, \psi)] + n \ln \psi(t) + n_0 \ln \bar{\psi}(0) - \ln n_0!}$$

$$= \int_{q(0)=n_0}^{q(t)=n} D(q,p) e^{\int dt [p\dot{q} + (e^{\beta} - 1)F_-(q) - (e^{-\beta} - 1)F_+(q)]}$$

$$= \int_{q(0)=n_0}^{q(t)=n} D(q,p) e^{\int dt [p\dot{q} - H(p,q)]}$$

... When $\mathcal{H}(t) =$

Exercise #8: (from exercise) "Classical Harmonic Oscillator"
(pg 677)

$$\hat{H}(x) = \frac{1}{2}(p^2 + q^2)$$

(from exercise) "Thermal Equilibrium"

$$p(x) = N e^{-H(x)/T}$$

(pg 677) "Two response Functions"

$$\text{Retarded} : C_q^+(t) = -i\theta(t) \langle [\hat{X}(t), \hat{X}(0)] \rangle$$

$$\text{Time-ordered} : C_q^T(t) = -i \langle T_t \hat{X}(t) \hat{X}(0) \rangle$$

Note: How are response functions possible from exercise?

Time is not position or momentum. The author on page 691 proposes, "The average observable position at time t " by a Kronecker delta, e.g. small error between time and position values."

$$\begin{aligned}
 \text{So, } C_q^\dagger(t) &= -i \Theta(t) \int_{-\infty}^{\infty} [X(t), X(0)] dt \\
 &= -i \Theta(t) \int_{-\infty}^{\infty} [\hat{X}(t) \hat{X}(0) - \hat{X}(0) \hat{X}(t)] dt \\
 &= -i \int_{-\infty}^{\infty} P(X) \cdot X \cdot dt \\
 &= -i \int_{-\infty}^{\infty} N e^{-H(X,t)/T} \cdot X \cdot dt \\
 &= -i \int N e^{-(p^2+q^2)/2T} dp dq \\
 &= -2\pi i T \cdot e \Big|_{-\infty}^{\infty}
 \end{aligned}$$

$$= -2\pi i T \left[\frac{1}{\infty} - \frac{1}{\infty} \right]$$

$$= 0$$

$$C_q^T(t) = -i \langle T_t \hat{X}(t) \hat{X}(0) \rangle$$

$$= -i \int_{-\infty}^{\infty} T_t P(x) \cdot x \, dt$$

$$= -i T_t \int_{-\infty}^{\infty} N e^{-\frac{(p^2+q^2)}{2T}} dp dq$$

$$= -i 2\pi T \cdot T_t N e^{-\frac{(p^2+q^2)}{2T}} \Big|_{-\infty}^{\infty}$$

$$= -i 2\pi T \cdot T_t N \left[\frac{1}{\infty} - \frac{1}{\infty} \right]$$

$$= 0$$

FDT of Linear Response:

$$\text{Im} C_q^{\dagger}(\omega) = \text{Im} C_q^{\dagger}(\omega) \coth(\beta\omega/2)$$

$$= \text{Im}(0) \coth(\beta\omega/2)$$

$$= 0$$

11.9.1 - Wigner Surmise:

(From problem) "Hamiltonian"

$$\hat{H} = \begin{pmatrix} H_1 & \Delta \\ \Delta & H_2 \end{pmatrix}$$

(from problem) "Probability"

$$P(H_1, H_2, \Delta) \propto e^{-\frac{1}{2} \text{tr } \hat{H}^2}$$

Joint Distribution:

$$P(E_+, E_-) \propto \int dH_1 dH_2 d\Delta e^{-(H_1^2 + H_2^2)/2}$$

$$= \int_0^\infty \int_0^\infty \int_0^\infty dH_1 dH_2 d\Delta \delta(E_+ - \bar{H} \pm \sqrt{\Delta^2 + \bar{H}^2} + \left(\frac{H_1 - H_2}{2}\right)^2) e^{-(E_+^2 + E_-^2)/2}$$
$$\times \delta(E_- - \bar{H} \pm \sqrt{\Delta^2 + \left(\frac{H_1 - H_2}{2}\right)^2}) e^{-(E_+^2 + E_-^2)/2}$$

$$= \int_0^\infty d(\delta H + H_2) dH_2 d\Delta \delta(E_+ - E_- - 2(\Delta^2 + \delta H^2)^{1/2}) e^{-(E_+^2 + E_-^2)/2}$$
$$\times e^{-(E_+^2 + E_-^2)/2}$$

$$\cong \int_0^\infty dr \cdot r \delta(E_+ - E_- - 2r) e^{-(E_+^2 + E_-^2)/2}$$

... when $r = (\Delta^2 + \delta H^2)^{1/2}$

$$\sim (E_+ - E_-) e^{-(E_+^2 + E_-^2)/2}$$

Energy Splitting:

$$P(s) \sim \int_{E_+ > E_-} dE_+ dE_- P(E_+, E_-) \delta(E_+ - E_- - s)$$

$$\sim s e^{-s^2/2}$$

$$= c_1 s e^{-c_2 s^2} \quad \dots \text{as } c_1 = 1 \text{ and } c_2 = 1/2$$

11.9.2-Ornstein-Uhlenbeck Process :

a) (11.83) "Ornstein-Uhlenbeck
Differential Equation"

$$\dot{v} + \gamma v = \frac{f(t)}{m}$$

When v = velocity
 $\dot{v}$ = acceleration

m = mass

$f(t)$ = Langevin
Force

Nonhomogenous First order Differential :

① Boundary conditions :

$$v(t=0) = v_0$$

② Variation by Parameters - Integrating Factor

$$e^{\gamma t} \dot{V} + e^{\gamma t} \cdot \gamma V = e^{\gamma t} \cdot \frac{f(t)}{m}$$

$$e^{\gamma t} \dot{V} + e^{\gamma t} \cdot \gamma V = \frac{d}{dt} (e^{\gamma t} \cdot V)$$

$$\frac{d}{dt} (e^{\gamma t} \cdot V) = e^{\gamma t} \cdot \frac{f(t)}{m}$$

$$V = e^{-\gamma t} \int e^{\gamma t} \cdot \frac{f(t)}{m} dt$$

$$= e^{-\gamma t} \left[e^{\gamma t} \cdot \frac{f(t)}{m} + c \right]$$

③ Solution by boundary conditions:

$$V(t=0) = e^{-\gamma \cdot 0} \left[e^{\gamma \cdot 0} \cdot \frac{f(0)}{m} + c \right]$$

$$= V_0, \quad \text{So } c = V_0$$

④ Exact Solution:

$$V(t) = e^{-\gamma t} \left[e^{\gamma t} \cdot \frac{f(t)}{m} + V_0 \right]$$

Book wrote solution by

$$V(t) = e^{-\gamma t} \left[V_0 + \int_0^t \frac{f(t')}{m} e^{\gamma t'} dt' \right]$$

Expectation $\langle v(t) \rangle$:

$$\dot{v} + \gamma v = \frac{f(t)}{m}$$

$$= 0$$

$$\int \frac{dv}{v} = -\gamma \int dt$$

$$v(t) = Ce^{-\gamma t} \quad @ \quad v(t=0) = v_0; C = v_0$$

$$v(t) = v_0 e^{-\gamma t}$$

Surprisingly, book denoted notation common to 'expectation values'. Although, 'steady-state' differential arrived to answer.

Expectation $\langle v^2(t) \rangle$:

$$(\dot{v} + \gamma v) 2(\dot{v} + \gamma v) = \frac{f(t)f(t')}{m^2}$$

Because unique times, t and t' , integrate nonhomogenous first order in steps.

$$V(t) = e^{-2\gamma t} \left[e^{2\gamma t} \frac{F(t)F(t')}{m^2} + v_0^2 \right]$$

Author added an integral to the solution.

$$V(t) = e^{-2\gamma t} \left[v_0^2 + \int_0^t dt' dt'' \frac{F(t')F(t'')}{m^2} e^{\gamma(t'+t'')} \right]$$

In the long-time limit, 'A' equates to $F(t')F(t'')$ and $2\gamma m t$.

b) (11.84) "Fokker Planck Equation"

$$\left[\frac{\partial}{\partial t} + \frac{\partial}{\partial v} a_1(v) - \frac{1}{2} \frac{\partial^2}{\partial v^2} a_2(v) \right] p(v, t) = 0$$

$\partial_t \langle v \rangle$ derived:

$$\partial_t \langle v \rangle = \frac{\partial}{\partial t} \langle v \rangle$$

$$= - \int dv v \partial_v (a_1 p) + \int dv \frac{1}{2} v \partial_v^2 (a_2 p)$$

$$= \int dv a_1 p - \frac{1}{2} \int dv \partial_v (a_2 p)$$

$$= \langle a_1 \rangle$$

$$\langle a_1 \rangle = \frac{d}{dt} \left[e^{-\gamma t} \left[v_0 + \int_0^t dt' \frac{F(t')}{m} e^{\gamma t'} \right] \right]$$

① Boundary Conditions:

$$v^2(t=0) = V_0^2$$

② Variation by Parameters - Integrating Factor:

$$2(e^{2\gamma t} \dot{v} + e^{2\gamma t} \cdot \gamma v^2) = e^{2\gamma t} \cdot \frac{f(t)f(t')}{m^2}$$

$$2(e^{2\gamma t} \dot{v} + e^{2\gamma t} \cdot \gamma v^2) = \frac{d}{dt} \left[e^{2\gamma t} \cdot v^2 \right]$$

$$\frac{d}{dt} \left[e^{2\gamma t} \cdot v^2 \right] = e^{2\gamma t} \cdot \frac{f(t)f(t')}{m^2}$$

$$v^2 = e^{-2\gamma t} \left[\int e^{2\gamma t} \cdot \frac{f(t)f(t')}{m^2} dt \right]$$

$$= e^{-2\gamma t} \left[e^{2\gamma t} \cdot \frac{f(t)f(t')}{m^2} + c \right]$$

③ Solution by boundary conditions:

$$v^2(t=0) = e^{-2\gamma \cdot 0} \left[e^{2\gamma \cdot 0} \cdot \frac{f(0)f(t')}{m^2} + c \right]$$

$$= V_0^2, \text{ so } c = V_0^2$$

④ Exact Solution:

$$= -\gamma v_0 \quad @ \quad t=0$$

$\partial_t \langle v^2 \rangle$ deriving:

$$\partial_t \langle v^2 \rangle = - \int dv \cdot v^2 \partial_v (a_1 P) + \int dv \frac{1}{2} v^2 \partial_v^2 (a_2 P)$$

$$= \int dv \cdot 2 \cdot v a_1 \cdot P - \int dv \cdot v \cdot \partial_v (a_2 P)$$

$$= \langle 2va_1 \rangle + \langle a_2 \rangle$$

$$\langle a_2 \rangle = \frac{\partial}{\partial t} \left[e^{-2\gamma t} \left[v_0^2 + \frac{A}{2\gamma m^2} (e^{2\gamma t} - 1) \right] \right] \cdot (-2\gamma v_0)$$

$$= -2\gamma v_0^2 + 2\gamma \frac{A}{m^2}$$

$$= 2 \left(\frac{\gamma T}{m} + \gamma v_0^2 \right)$$

Generating Function:

(from problem) "Fourier Integral"

$$g(k, t) = \int dv e^{-ikv} \cdot P(v, t)$$

$$\left[\frac{\partial}{\partial t} + \frac{\partial}{\partial v} a_1 \langle v \rangle - \frac{1}{2} \frac{\partial^2}{\partial v^2} a_2 \langle v \rangle \right] P(v, t)$$

$$= \left[\frac{\partial}{\partial t} - \frac{\partial}{\partial v} \gamma v_0 - \frac{\gamma T}{m} \frac{\partial^2}{\partial v^2} \right] P(v, t)$$

$$= \left[\frac{\partial}{\partial t} + \gamma k i \frac{\partial}{\partial k} + \frac{\gamma T}{m} k^2 \right] g(k, t)$$

$$= 0$$

$\leftarrow$ left from

Fourier Transform,

c)

(11.85) "Generating Function"

$$\left[\frac{\partial}{\partial t} + \gamma k \frac{\partial}{\partial k} + \frac{\gamma T}{m} k^2 \right] g(k, t) = 0$$

11.9.4 - Ornstein-Uhlenbeck process revisited:

(From problem) "Overdamped Oscillator"

$$\ddot{q} + \frac{\omega_0^2}{\gamma} q = 0$$

(from problem) "Path Integral"

$$P(q_f, t | q_i, 0) = \int_{q(t)=q_f: q(0)=q_i} D[q, p] e^{-S[q, p]}$$

$$\text{where } S = - \int_0^t dt (p \dot{q} - H)$$

$$H = -\alpha p q - \frac{A}{2} p^2$$

$$\alpha = \omega_0^2 / \gamma$$

a) Hamiltonian Solutions:

$$\text{velocity} = \dot{q}$$

$$= \frac{\partial H}{\partial p}$$

$$= -\alpha q - A p$$

$$\text{Acceleration} = \dot{p}$$

$$= -\frac{\partial H}{\partial q}$$

$$= \alpha p$$

$$\text{Finally, } p(t) = p_0 e^{\alpha t}$$

$$q(t) = c e^{-\alpha t} - \frac{A p_0}{2\alpha} e^{\alpha t}$$

① Boundary Conditions:

$$q(t=0) = q_i \text{ and } q(t) = q_f$$

② Solutions by Boundary Conditions:

$$C: q(t=0) = c e^{-\alpha \cdot 0} - \frac{A p_0}{2 \cdot \alpha} e^{+\alpha \cdot 0}$$

$$= c - \frac{A p_0}{2 \cdot \alpha}$$

$$= q_i, \text{ So } c = q_i + \frac{A p_0}{2\alpha}$$

$$A: q(t) = c e^{-\alpha t} - \frac{A p_0}{2\alpha} e^{\alpha t}$$

$$= \left(q_i + \frac{A p_0}{2\alpha} \right) e^{-\alpha \cdot t} - \frac{A p_0}{2\alpha} e^{\alpha t}$$

$$= q_f, \text{ So,}$$

$$A = \frac{2\alpha}{p_0} \frac{(q_f e^{\alpha t} - q_i)}{(1 - e^{2\alpha t})}$$

③ Exact Solution:

$$P(t) = P_0 e^{\alpha t}$$

$$q(t) = \left[\frac{q_i - q_F e^{-\alpha t}}{1 - e^{-2\alpha t}} \right] (e^{-\alpha t} + e^{\alpha t})$$

Classical Action:

(from problem) "Action"

$$S = - \int_0^t dt' (p \dot{q} - H)$$

$$= - \int_0^t dt' (p \dot{q} + \alpha p q + \frac{A}{2} p^2)$$

$$= \frac{\alpha}{A} \frac{(q_i - q_F e^{\alpha t})^2}{e^{2\alpha t} - 1}$$

b) (3.27) "Transition Amplitude For

Semiclassical Approximation"

$$\langle q_F | e^{-\frac{i}{\hbar} \frac{p^2}{2m} t} | q_i \rangle \Theta(t) = \left(\frac{m}{2\pi i \hbar t} \right)^{1/2} e^{\frac{i}{\hbar} \frac{m (q_F - q_i)^2}{2t}} \cdot \Theta(t)$$

$$\langle q_F | e^{-\frac{i}{\hbar} H t} | q_i \rangle \simeq \det \left(\frac{i}{2\pi \hbar} \frac{\partial^2 S[q_c]}{\partial q_i \partial q_F} \right)^{1/2} e^{\frac{i}{\hbar} S[q_i]}$$

Hamiltonian:

(part a) "Hamiltonian"

$$H = -\alpha p q - \frac{A}{2} p^2 \text{ where } \alpha = \frac{\omega_0^2}{8}$$

If p was an operator,

$$\begin{aligned} \text{then } H &= -\alpha [p, q] - \frac{A}{2} p^2 \\ &= \overset{\text{Ansatz } \frac{11}{10}}{\uparrow} \\ &= -\alpha (q \cdot p - p \cdot q + 1) - \frac{A}{2} p^2 \\ &= \overset{\uparrow}{[A, B] = AB - BA} \\ &= 1 \\ &= -\alpha \frac{(qp - pq)}{2} - \frac{\alpha}{2} - \frac{A}{2} p^2 \end{aligned}$$

Action:

$$\langle q_F | e^{-iHt/\hbar} | q_i \rangle = \det \left(\frac{i}{2\pi\hbar} \frac{\partial^2 S[q_c]}{\partial q_i \partial q_f} \right)^{1/2} e^{\frac{i}{\hbar} (S[q_c] - \frac{\alpha}{2})}$$

Focke-Planck Equation:

() "Focke-Planck Equation"

$$(\partial_t + \alpha \partial_q q + \frac{A}{2} \partial_q^2) P = 0$$

$$= \frac{\alpha}{2} (\partial_q P + P \partial_q) + \frac{A}{2} \partial_q^2 + \frac{\alpha}{2}$$

$$= \frac{A}{2} \left(\partial_q + \frac{\alpha}{A} q \right)^2 - \left(\frac{\alpha^2}{2A} \right) q^2 + \frac{\alpha}{2}$$

C) (11.96) "Transition Amplitude"

$$\frac{1}{\sqrt{2\pi}} \sqrt{-\frac{\partial^2 S_{cl}}{\partial q_i \partial q_f}} = -\frac{1}{\sqrt{2\pi}} \sqrt{-\frac{\partial^2}{\partial q_i \partial q_f} \left(\frac{\alpha}{A} \frac{(q_i - q_f e^{\alpha t})^2}{e^{2\alpha t} - 1} \right)}$$

$$= -\frac{1}{\sqrt{2\pi}} \sqrt{-\frac{\partial}{\partial q_f} \cdot \frac{\alpha}{A} \cdot \frac{2(q_i - q_f e^{\alpha t})}{e^{2\alpha t} - 1}}$$

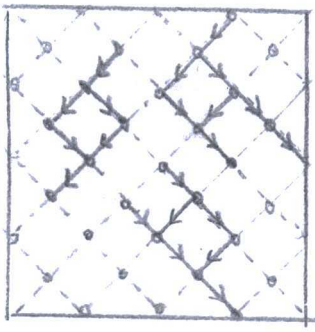
$$= -\frac{1}{\sqrt{2\pi}} \sqrt{-\frac{\alpha}{A} \frac{2(-e^{\alpha t})}{e^{2\alpha t} - 1}}$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{2\alpha}{A} \right)^{1/2} \frac{e^{-\alpha t/2}}{(1 - e^{-2\alpha t})^{1/2}}$$

$$P(q_f, t | q_i, 0) = -\frac{1}{\sqrt{2\pi}} \sqrt{-\frac{\partial^2 S_{cl}}{\partial q_i \partial q_f}} e^{\frac{\alpha}{A} \frac{(q_i - q_f e^{\alpha t})^2}{e^{2\alpha t} - 1} + \frac{\alpha}{2} t}$$

$$= -\frac{1}{\sqrt{2\pi}} \left(\frac{2\alpha}{A} \right)^{1/2} \frac{e^{-\frac{\alpha}{A} \frac{(q_i - q_f e^{\alpha t})^2}{e^{2\alpha t} - 1}}}{(1 - e^{-2\alpha t})^{1/2}}$$

Problem 11.9.5 - Directed Percolation:



Personal Note:
Path-dependent
energy
with

- a) Jackson - Classical Electrodynamics extended path dependent energy by interior points in a volume. Transmission lines, magnets, and conductors are common in simulations with points, vertices and edges. The 2D-model became a volume for students. Where distance between points was Green's function or Potential. Reference: Chapters 1/2 - Jackson.

(11.79) "Field Theory Representation of Directed Percolation"

$$S[\phi, \psi] = \int dx \left[\underbrace{\psi(\partial_t - D\partial^2 - \pi)\phi}_{\text{"Initial Action } [S_0]"} + \underbrace{K\psi\phi(\phi - \psi)}_{\text{"Interaction } [S_{int}]"} \right]$$

"Initial
Action $[S_0]$ "

"Interaction
 $[S_{int}]$ "

(one dimensional)

$$S_{int} = K \int dx \psi \phi (\psi - \psi)$$

(two dimensional)

$$\begin{aligned} S_{int} &= K^2 \int dx \cdot dx' \psi_q \phi_{q'} (\phi_{-q-q'} - \psi_{-q-q'}) \\ &= K^2 \int b dq b dq' \psi_q \phi_{q'} (\phi_{-q-q'} - \psi_{-q-q'}) \\ &= K^2 b^2 \int dq dq' \psi_q \phi_{q'} (\phi_{-q-q'} - \psi_{-q-q'}) \end{aligned}$$

(11.37) "Green Function"

$$\langle \phi_q \psi_{q'} \rangle = g_q^+ \delta_{q, -q'} \quad (g_q^+)^{-1} = -i\omega + DK^2 - \tau$$

$$g_{-q}^- = g_q^+$$

$$S_{int} = -2K^2 \int (g_{q-q'} g_{q'} + g_{q+q'} g_{q'}) dq$$

$$= 2K^2 \int \frac{d^d k'}{(2\pi)^d} \frac{1}{-i\omega + DK'^2 + D(K-K')^2 - 2\tau}$$

Square Root Expansion 

Lowest Order Correction:

$$\frac{-1}{3!} \langle S_{int}^3 \rangle = \frac{2^4 \cdot 3}{3!} K^3 \int dq dq' dq'' \phi_q^2 \phi_{q'}^2 \phi_{q''}^2 \times$$

$$\left(-\phi_{-q-q'-q''}^2 + \phi_{q-q'-q''}^2 \right) \times$$

$$\int dP g_P^2 g_P$$

$$= S_{int}[\phi, \psi] \circ \delta_K$$

$$\text{where } \delta_K = -(2K)^3 \int dq g_q^2 g_{-q}$$

$$= -2K^3 \int \frac{d^d k'}{(2\pi)^d} \frac{1}{(DK^2 - \tau)^2}$$

b) Momentum Integral:

(11.88) "Self-Energy correction"

$$\Sigma_q = 2K^2 \int \frac{d^d k'}{(2\pi)^d} \frac{1}{-i\omega + DK'^2 + D(k-k')^2 - 2\tau}$$

$$= -2K^2 \int \frac{d^d k'}{(2\pi)^d} \frac{1}{(i\omega + 2\tau) - DK'^2 + D(k-k')^2}$$